Element C: Presentation and Justification of Solution Design Requirements

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students’ ability to rest.

People who have insects in their rooms need to remove the insects in order to sleep safely. The new solution must meet specific requirements to be effective. It should be able to trap and remove insects from one area in a way that is quick and easy to manage. The setup must be small and compact so that it does not take up too much space in the bedroom. Ideally, the solution should take less than ten minutes to prepare or reset. The solution used to eliminate the insects must be safe for people and pets and should not harm the environment. Although appearance is not the top priority, the solution should be visually appealing so as not to stand out in the room. Finally, it must include features that stop insects from returning to the area. A more advanced feature, to repel or eliminate insects at their source, would improve long-term effectiveness and meet the students' needs. The prioritized design requirements are:

1. **Neutralization Efficiency -** The design must be able to catch a variety of insects, and catch enough insects for there to be a noticeable difference in insect populations. (high)
2. **Smaller than 14 cubic inches -** The design must be small enough to fit on Student E’s shelf. (14”x14”x14”). (high)
3. **Quick Assembly and Application -** The design should prioritize ease of setup and use. This means the insect-catching mechanism should be straightforward to assemble within 10 minutes, without requiring specialized tools or extensive instructions. Similarly, applying or deploying the device should be quick and

simple, allowing for convenient placement and activation in areas where insects are a concern. (high)

4. **Ethical design -** The design must operate with the well-being of all living things and the environment in mind. Specifically, it does not harm pets, humans, or the environment. This implies using non-toxic methods and ensuring the device poses no risk to unintended targets. (high)
5. **Aesthetically Pleasing Design -** The final product should have a nice aesthetic, allowing it to integrate reasonably well into different environments without being intrusive or unsightly. The design should aim for a visually acceptable form and appearance. (medium)

Student E agreed with the design criteria and gave reasonable feedback on feasible ideas. The list has been altered slightly to accommodate Student E's feedback. Student E asked to add a size requirement, so that the design would fit on his shelf. He also asked that it not include a chemical lure for insects.